

SQL PROJECT ON ONLINE BOOK STORE

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Abstract

This project analyse a bookstore dataset using SQL queries across three tables: **Books, Orders, and Customers**. The objective is to evaluate sales performance, customer spending behavior, genre-wise revenue distribution, and identify top customers. The findings support data-driven decision-making for marketing strategies, inventory management, and customer relationship optimization.

ER DIAGRAM

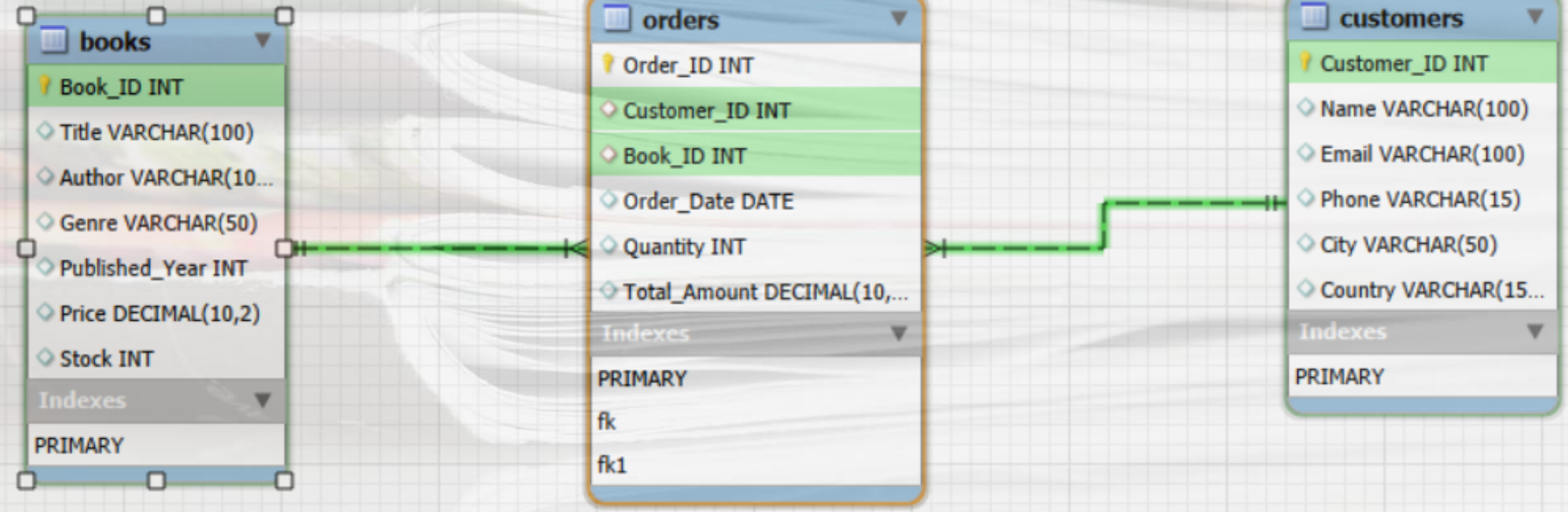

books
Book_ID INT
Title VARCHAR(100)
Author VARCHAR(10...
Genre VARCHAR(50)
Published_Year INT
Price DECIMAL(10,2)
Stock INT
Indexes
PRIMARY
orders
Order_ID INT
Customer_ID INT
Book_ID INT
Order_Date DATE
Quantity INT
Total_Amount DECIMAL(10,...
Indexes
PRIMARY
fk
fk1
customers
Customer_ID INT
Name VARCHAR(100)
Email VARCHAR(100)
Phone VARCHAR(15)
City VARCHAR(50)
Country VARCHAR(15...
Indexes
PRIMARY

TABLE STRUCTURE

1) Books Table

Result Grid | Filter Rows:

Field	Type	Null	Key
Book_ID	int	NO	PRI
Title	varchar(100)	YES	
Author	varchar(100)	YES	
Genre	varchar(50)	YES	
Published_Year	int	YES	
Price	decimal(10,2)	YES	
Stock	int	YES	

In this Table, The books column defines the title of the books, Author, genre, publish year, price and stock, Book_ID as a Primary Key

TABLE STRUCTURE

2) Customers Table

Field	Type	Null	Key
Customer_ID	int	NO	PRI
Name	varchar(100)	YES	
Email	varchar(100)	YES	
Phone	varchar(15)	YES	
City	varchar(50)	YES	
Country	varchar(150)	YES	

In this Table Customers details is showing , like name, email, phone, city ,country. Customers_ID as a Primery Key

TABLE STRUCTURE

3) Orders Table

Field	Type	Null	Key
Order_ID	int	NO	PRI
Customer_ID	int	YES	MUL
Book_ID	int	YES	MUL
Order_Date	date	YES	
Quantity	int	YES	
Total_Amount	decimal(10,2)	YES	

In this Orders Table Showing Order_ID, Customer_ID, Books_ID, Order_date, Quantity, Total_amount
Primery Key define as Order_ID, And Foraign key as Customer_ID & Books_ID

Assumptions

1. Each `Customer_ID`, `Book_ID`, and `Order_ID` is unique and correctly mapped.
2. `Total_Amount` = `Quantity × Price`; missing values are treated as zero.
3. Every order record represents a confirmed purchase (no cancellations included).
4. Each book belongs to exactly one genre; genre names are consistent.
5. Each customer is linked to one city only.
6. Books with no matching orders are considered “never ordered.”

Exploring book sales and customer trends using SQL analytics.

❖ Retrieve all books in the "Fiction" genre:

```
select * from books where genre = "Fiction";
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79
29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.12	100
31	Implemented encompassing conglomeration	Melissa Taylor	Fiction	2010	21.23	44
39	Optimized national process improvement	Megan Goodwin	Fiction	1978	10.99	42
40	Adaptive didactic interface	Natalie Gonzalez	Fiction	1923	25.97	94
47	Reverse-engineered directional conglomeration	John Christian	Fiction	2006	20.37	90
62	Re-contextualized real-time strategy	Nicole Lynch	Fiction	1953	26.34	23
63	Polarized heuristic database	Franklin Mack	Fiction	1989	22.38	56
100	Synchronized client-server service-desk	James Alvarado	Fiction	1906	49.89	29

Insight :
“Fiction books show variety in price and stock, so demand is strong and inventory planning is important.”

Exploring book sales and customer trends using SQL analytics.

❖ Find books published after the year 1950:

```
select * from books where Published_Year > 1950;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84
9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.33	70
10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.63	25
11	Secured zero tolerance time-frame	Denise Barnes	Fantasy	1998	35.95	10
12	Polarized optimal array	Destiny Scott	Non-Fiction	1989	27.43	63
15	User-friendly motivating strategy	Keith Smith	Non-Fiction	1997	23.83	58
17	Reduced secondary core	Benjamin Peters	Fantasy	1966	5.37	45

Insight :
“Books published after 1950 show variety in genres, higher prices for recent titles, and mixed stock levels — pointing to strong demand for new releases and surplus in older ones.”

Exploring book sales and customer trends using SQL analytics.

- List all customers from the Canada:

```
select * from Customers where country = "Canada";
```

Customer_ID	Name	Email	Phone	City	Country
38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
468	David Hart	stokesrebecca@gmail.com	1234568358	Thompsonfurt	Canada
NULL	NULL	NULL	NULL	NULL	NULL

Insight:

“Canadian customers are present across multiple cities, showing a geographically diverse base.”

Exploring book sales and customer trends using SQL analytics.

- Show orders placed in November 2023:

```
select * from Orders where order_date between '2023-11-01' and '2023-11-30';
```

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
4	433	343	2023-11-25	7	301.21
19	496	60	2023-11-17	9	316.26
75	291	375	2023-11-30	5	170.75
132	469	333	2023-11-22	7	194.32
137	474	471	2023-11-25	8	363.04
163	207	384	2023-11-23	3	101.76
182	129	293	2023-11-01	7	125.51
200	313	303	2023-11-23	1	6.57
213	325	447	2023-11-17	7	253.75
231	22	384	2023-11-11	1	33.92
245	386	97	2023-11-01	9	411.66



Exploring book sales and customer trends using SQL analytics.

- Retrieve the total stock of books available:

```
select sum(stock) as Books_Stock from Books;
```

Books_Stock
25056

Insights:

“The bookstore has a total inventory of 25,056 books available.”

Exploring book sales and customer trends using SQL analytics.

- Find the details of the most expensive book:

```
select * from books  order by price desc limit 1;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
340	Proactive system-worthy orchestration	Robert Scott	Mystery	1907	49.98	88
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Insight:

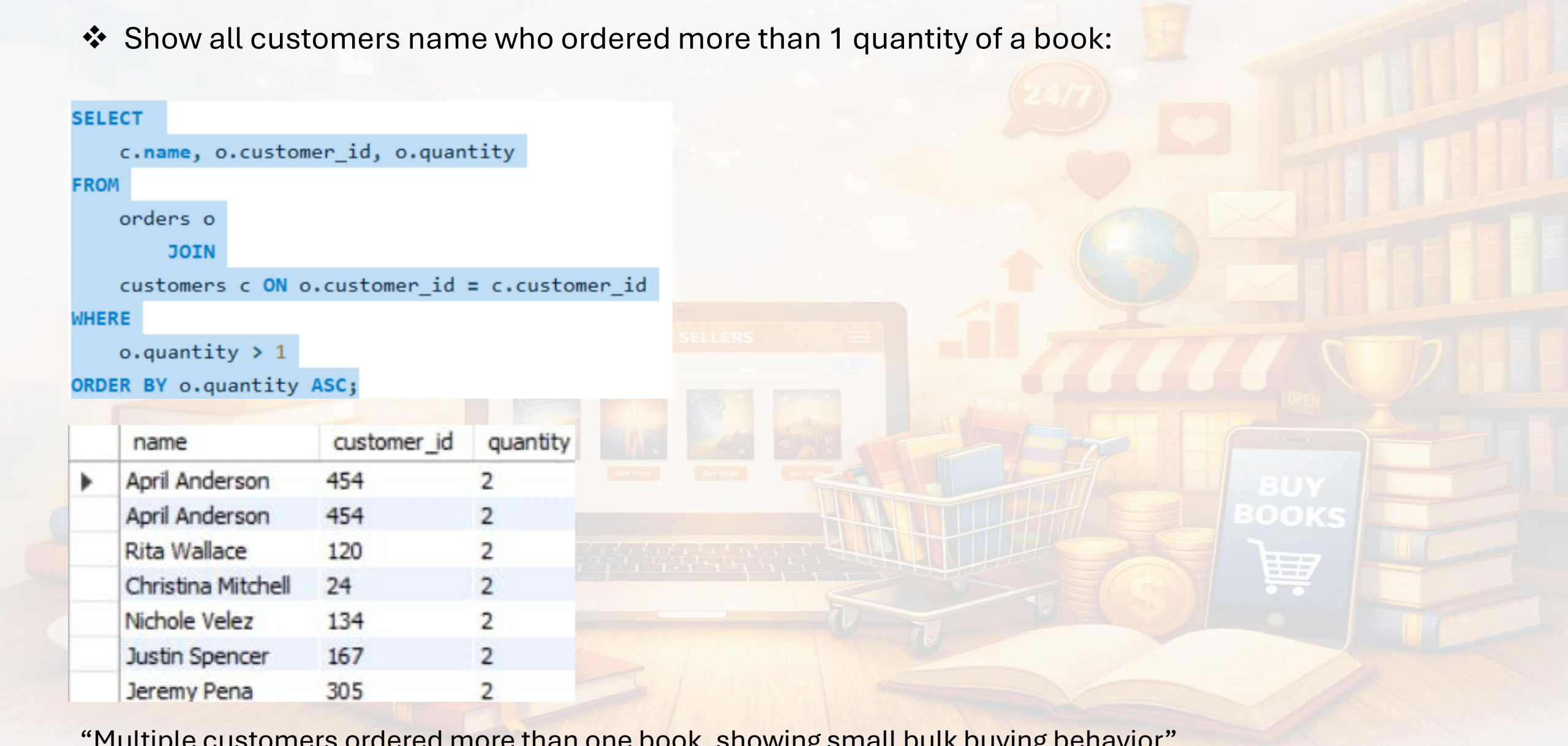
"The Mystery book 'Proactive system-worthy orchestration' is the highest-priced title, offering premium value with good stock availability."

Exploring book sales and customer trends using SQL analytics.

- Show all customers name who ordered more than 1 quantity of a book:

```
SELECT
    c.name, o.customer_id, o.quantity
FROM
    orders o
        JOIN
    customers c ON o.customer_id = c.customer_id
WHERE
    o.quantity > 1
ORDER BY o.quantity ASC;
```

	name	customer_id	quantity
▶	April Anderson	454	2
	April Anderson	454	2
	Rita Wallace	120	2
	Christina Mitchell	24	2
	Nichole Velez	134	2
	Justin Spencer	167	2
	Jeremy Pena	305	2



“Multiple customers ordered more than one book, showing small bulk buying behavior”

Exploring book sales and customer trends using SQL analytics.

- Retrieve all orders where the total amount exceeds $20:

```
select * from orders where Total_Amount > 20 order by Total_Amount ;
```

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
307	368	133	2023-11-17	1	20.96
100	207	63	2023-07-14	1	22.38
260	112	170	2024-05-07	2	22.56
319	172	296	2024-06-27	2	22.98
345	325	253	2023-07-04	2	23.32
494	117	127	2024-11-23	2	23.32
288	6	128	2023-11-13	1	24.04

Insight:

“Orders above $20 reflect small, steady purchases with customers typically buying 1–2 books.”

Exploring book sales and customer trends using SQL analytics.

- List all genres available in the Books table:

```
select distinct(Genre) from books;
```

Genre
Biography
Fantasy
Non-Fiction
Fiction
Romance
Science Fiction
Mystery

Insight:

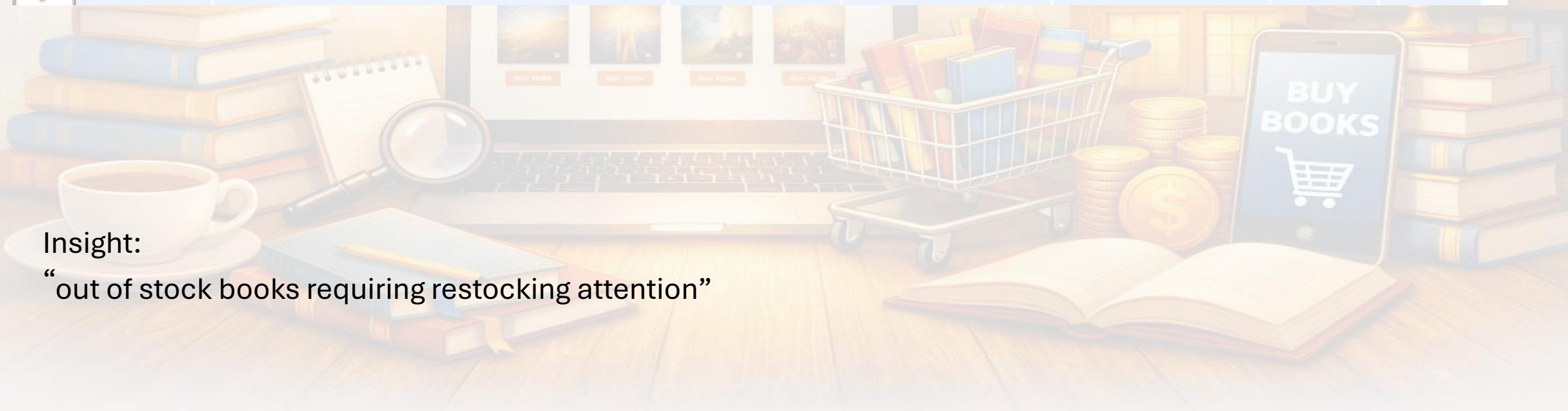
“The Books table offers 7 distinct genres, ensuring a wide variety for different reader preferences.”

Exploring book sales and customer trends using SQL analytics.

- Find the book with the lowest stock:

```
select * from books order by stock asc limit 1;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
44	Networked systemic implementation	Ryan Frank	Science Fiction	1965	13.55	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL



Insight:
“out of stock books requiring restocking attention”

Exploring book sales and customer trends using SQL analytics.

❖ Calculate the total revenue generated from all orders:

```
select sum( Total_Amount) As Total_Revenue from Orders;
```

	Total_Revenue
▶	75628.66

Insight:

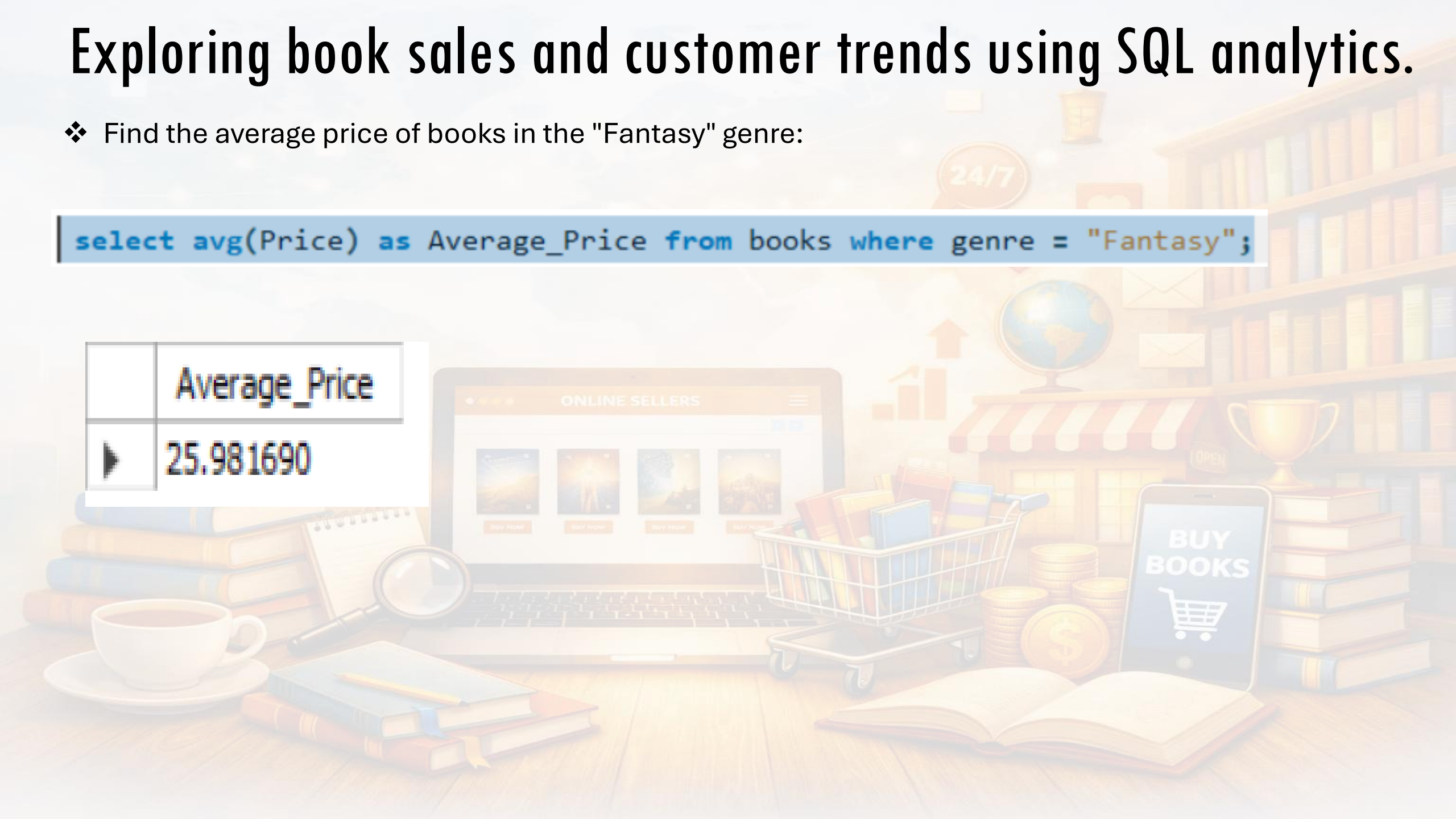
“Total revenue from all orders is 75,628.66, reflecting overall business sales performance”

Exploring book sales and customer trends using SQL analytics.

- Find the average price of books in the "Fantasy" genre:

```sql
select avg(Price) as Average_Price from books where genre = "Fantasy";
```

	Average_Price
▶	25.981690



Exploring book sales and customer trends using SQL analytics.

❖ Show the top 3 most expensive books of 'Fantasy' Genre :

```
select * from books  where  genre = 'Fantasy' order by Price desc limit 3;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41
462	Innovative 3rdgeneration database	Allison Contreras	Fantasy	1988	49.23	62
238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72

Insight:

“The top 3 Fantasy books are premium-priced, ranging from 48.97 to 49.90, highlighting high-value offerings in the genre”

Exploring book sales and customer trends using SQL analytics.

❖ Retrieve the total number of books sold for each genre:

```
select sum(o.Quantity) as total_orders , b.genre from orders o join books b on o.book_id = b.book_id group by b.genre ;
```

total_orders	genre
285	Biography
446	Fantasy
447	Science Fiction
504	Mystery
439	Romance
351	Non-Fiction
225	Fiction

Insight:

“Mystery leads sales with 504 books, while Fiction has the lowest at 225.”

Exploring book sales and customer trends using SQL analytics.

❖ Retrieve the total quantity of books sold by each author:

```
select sum( o.quantity) as Total_sold, b.author from orders o join books b on o.book_id = b.book_id group by b.author order by Total_sold  desc ;
```

Total_sold	author
28	Patrick Contreras
27	Melissa Taylor
24	Emily James
24	Thomas Trujillo
23	Erica Parker
23	Sheena Harris
23	Valerie Moore
23	Ellen Doyle

Insight:

“Patrick Contreras leads with 28 books sold, followed closely by Melissa Taylor and others in the mid-20s range.”

Exploring book sales and customer trends using SQL analytics.

- List customers who have placed at least 2 orders:

```
select count(o.Order_id) as order_quantity,c.name, o.customer_id from orders o join customers c on
c.customer_id = o.customer_id group by o.customer_id, c.name having count(o.Order_id) >=2;
```

order_quantity	name	customer_id
2	Crystal Clements	2
2	Stephen Vasquez	6
2	Matthew Johnson	8
2	Kristine Kim	13
2	John Wood	14
2	Vanessa Gaines	15
3	Stacey Flores	16

Insight:

Many customers placed at least 2 orders, showing repeat purchase behavior and potential loyalty.”

Exploring book sales and customer trends using SQL analytics.

- Find the most frequently ordered book:

```
select b.Title, b.Book_id,count(o.order_id) from books b join orders o
on b.book_id = o.book_id group by b.Title, o.book_id order by count(o.order_id) desc limit 1;
```

	Title	Book_id	count(o.order_id)
▶	Implemented encompassing conglomeration	31	4

Insight:

The book ‘Implemented encompassing conglomeration’ is the most ordered title, showing strong customer demand

Exploring book sales and customer trends using SQL analytics.

- List the cities where customers who spent over $30 are located:

```
select distinct c.city ,o.total_amount from customers c join orders o on o.customer_id = c.customer_id  where total_amount >30 ;
```

	city	total_amount
▶	Lake Paul	188.56
	North Keith	216.60
	Kelseyfort	85.50
	East David	301.21
	Richardsonville	136.36
	Ramosstad	249.40
	Rogersborough	82.92

Insight:

High-spending customers are located across several cities, with East David and Ramosstad leading in total spend

Exploring book sales and customer trends using SQL analytics.

❖ Find the customer who spent the most on orders:

```
select o.customer_id,c.name, sum(o.total_amount) as total_spent from customers c join orders o on o.customer_id = c.customer_id
group by o.customer_id, c.name order by total_spent desc limit 1;
```

customer_id	name	total_spent
457	Kim Turner	1398.90

Insights:

Kim Turner is the top spender, driving the highest revenue and should be prioritized for rewards.

Exploring book sales and customer trends using SQL analytics.

- Calculate the stock remaining after fulfilling all orders:

```
select b.book_id, b.title, b.stock, coalesce(sum(o.quantity),0) as order_quantity, b.stock - coalesce(sum(o.quantity),0) as Remaining_quantity
from books b join orders o on b.book_id = o.book_id group by b.book_id, b.title order by Remaining_quantity desc;
```

book_id	title	stock	order_quantity	Remaining_quantity
475	Stand-alone zero administration emulation	100	1	99
1	Configurable modular throughput	100	3	97
29	Quality-focused multi-tasking challenge	100	4	96
254	Team-oriented bandwidth-monitored project	100	4	96
368	Stand-alone logistical installation	97	2	95
370	Synergistic user-facing frame	97	2	95
148	Upgradable contextually-based hierarchy	97	3	94

Insights:

Orders are low, so most books have plenty of stock remaining; overall inventory is secure

Exploring book sales and customer trends using SQL analytics.

- Find books priced above the average price.

```
SELECT Title, Price FROM Books WHERE Price > (SELECT AVG(Price) FROM Books);
```

	Title	Price
▶	Persevering reciprocal knowledge user	35.80
	Customizable 24hour product	43.52
	Open-architected exuding structure	43.63
	Persistent local encoding	48.99
	Secured zero tolerance time-frame	35.95
	Polarized optimal array	27.43
	Adaptive 4thgeneration concept	39.47



Insights:

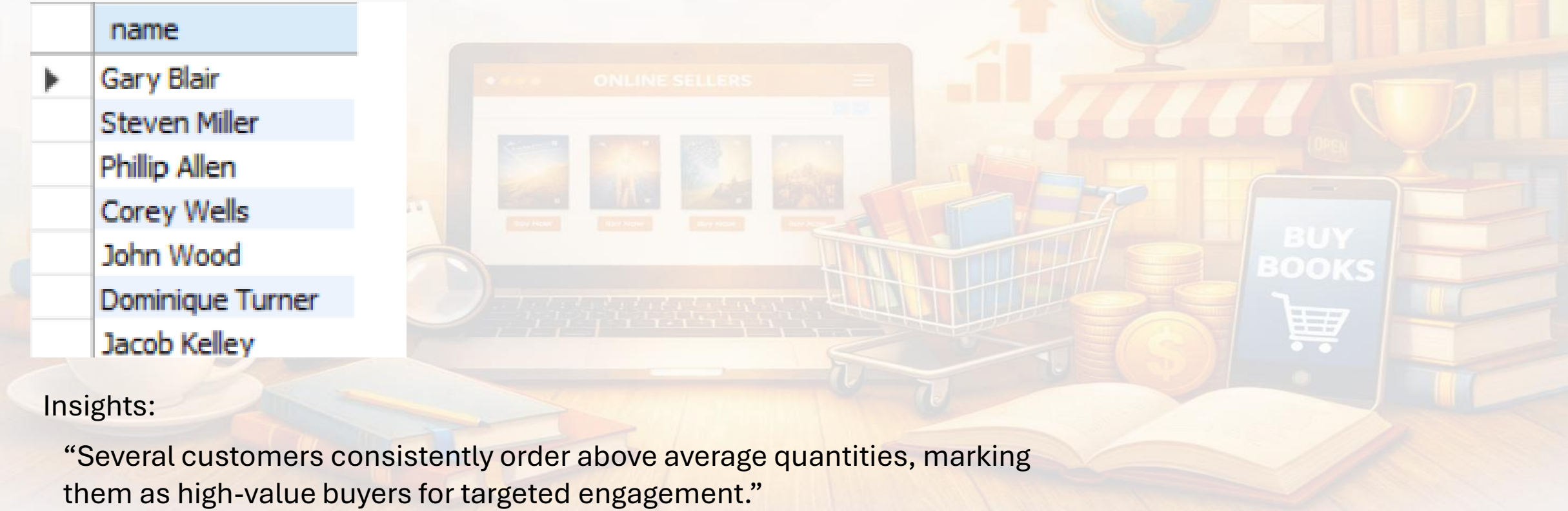
Above-average priced books highlight premium titles with higher margins

Exploring book sales and customer trends using SQL analytics.

- Find customers who ordered more than average quantity.

```
select distinct c.name from customers c join orders o on c.Customer_ID = o.Customer_ID where o.Quantity > ( select avg(Quantity) from orders);
```

name
Gary Blair
Steven Miller
Phillip Allen
Corey Wells
John Wood
Dominique Turner
Jacob Kelley



Insights:

"Several customers consistently order above average quantities, marking them as high-value buyers for targeted engagement."

Exploring book sales and customer trends using SQL analytics.

- Create a view showing customer name + total purchase.

```
create view Customers_sale as select c.name, sum(o.Total_amount) as total_purchase from
customers c join orders o on c.Customer_ID=o.Customer_ID group by c.name;
```

```
select * from Customers_sale;
```

name	total_purchase
Gary Blair	231.68
Steven Miller	286.92
Phillip Allen	157.80
Corey Wells	301.21
John Wood	383.06
Shane Chang	249.40
Dominique Turner	480.42



📌 Conclusion

•This SQL analysis highlights clear patterns in book sales, customer behavior, and inventory trends across the bookstore dataset.

•Revenue generation is concentrated in top genres like Mystery and high-spending customers such as Kim Turner, while some genres (like Fiction) show weaker demand.

•Insights reveal opportunities to improve performance by restocking out-of-stock titles, rewarding loyal repeat buyers, and promoting underperforming authors/genres.

•By acting on these findings, the bookstore can optimize inventory planning, strengthen marketing strategies, and enhance customer loyalty programs, ultimately driving sustainable growth and higher profitability.